

CONCEPTS SECTION LEARNING OBJECTIVES

1) Engineering Design Process

- Formulating the problem,
- Implementing a solution,
- Using Quality Principles.

2) Working in Teams

- Team dynamics,
- Team communication,
- Social norms,
- Conflict management.

3) Engineering as a Profession

- The profession,
- Selection of a major,
- Professional ethics.

4) Planning for Success

- Learning to learn,
- Assessment of progress,
- Time management.

In general; this package of materials is intended to help you get started in the **Concepts** Session of GE 211. The package contains material addressing

1. Concepts Expectations,
2. Quizzes, Assignments (with schedule)
3. Assessment,
4. Resubmission of Work,
5. Quiz information
6. Concepts Grade, and
7. Academic Integrity.
8. How to appeal for correct answers
9. Engineering Journal

1- CONCEPTS SESSION EXPECTATIONS

The expectations for the Concepts Session of GE 211 are listed below. **Meeting Customer Expectations is a characteristic of Self-Regulated Learners** (one of the 5 Course Objectives for GE 211.) We expect that each student will:

- Complete the assigned readings before class (see the GE 211 Course Calendar).
- Attend and be on time for the scheduled class meetings.
- Actively participate in all learning activities and tasks.
- Treat your classmates with respect
- **Avoid disruptive behaviors** in class (e.g., side conversations, sleeping, working on homework assignments for other classes, etc.) (see Section 4 of the First Day handout)
- Sign the Concepts attendance sheet before each class begins
- Meet or Exceed Expectations on all the work products (i.e., quizzes, etc.).
- Improve all permitted work that is assessed as Needs Improvement, No Credible Effort, or No Submission.
- Conduct/submit the additional work assigned by the end of the semester (Usually named Defect removal) to remove ONE defect (optional).
- **Always review your grades (as you can calculate the grades you deserve based on the announced instructions, grades, ... etc.).** Afterward; If there is a difference between the announced and calculated grades; you will need to have your original work to verify your actual grade. (25 students $\times$ 10 quizzes + Journals + Homework .. etc. = a lot of opportunity for data entry errors!),

2- QUIZZES AND CONCEPTS ASSIGNMENTS

The Concepts section of the course is intended to introduce you to the major ideas that you will use in the Laboratory section, where you will practice using them on actual design projects. We use quizzes given in nearly every session to ascertain whether you have understood the material in your pre-class reading assignments. In addition, we encourage students to write brief reflective journal entries, mainly for those who get NCE/NS in the first submission, to 1) further solidify and reinforce their own understanding, and 2) demonstrate that improved understanding for an improved Quiz grade. (Well-done Journal Entries can lead to an improved Concepts Grade, section on Assessment). The Concepts assignments and information can be found in the GE 211 materials provided to you. You will want to print and keep these assignments.

The assignment(s) and quiz schedule is shown in the hereafter in Table 1. You will want to study this schedule and make sure you are prepared for each Concepts Session.

All work products must be submitted to the instructor, during the **first 10 minutes** of the due session. In the case of online classrooms, the instructor will inform you of the way to submit your work. *Usually, you get this information through a blackBoard announcement and/or by mail.* More about quizzes is given in section 5 of this file.

3- ASSESSMENT OF WORK PRODUCTS

There are two types of work assessed in Concepts: your quizzes and your written material. As is true for all work in GE 211, your work will be assessed as Exceeding Expectations, Meeting Expectations, Needing Improvement, No Credible Effort, or Not Submitted. Your quiz assessment is based on the number of questions you answered correctly. The relationship between correct answers, journals, and expectations is shown in Table 2.

All written material will be assessed using checklists. You can print/use all the checklists for all the written assignments from the course material provided to you. The checklists you are going to use must be the correct one and must belong to the provided set for the current semester. The use of an inappropriate checklist **will be considered as no checklist is submitted.**

You are expected to include a checklist with each written assignment you submit (if applicable) . it will help you to understand how the work's grade is calculated. You are expected to follow the instructions on each checklist concerning the need to self-assess (i.e., fill out) the checklist before submitting (most work is self-assessed). The **checklist** should be the **top page** of all written assignments.

4- RESUBMISSION OF WORK

- As indicated in the white first-day materials, you may resubmit only permitted work (e.g., quiz correction/makeup) which is initially assessed as NI, NCE, or NS for reassessment.
- Table 1, in yellow Concept's first day materials, gives the due date for the resubmission.
- All re-submissions of work must include the **Resubmission** Checklist (if applicable), which explains the details of how to prepare the resubmission. The Resubmission Checklist can be found in the GE 211 materials.
- You will get one Self Regulation laps if re-submission is delivered within one week from its due date. **No re-submission** is accepted after one week of its due date.
- With the exception of your Engineering Journals; **quiz** that is initially assessed as NI, NCE or NS may be resubmitted for re-assessment.
- The dates for resubmission are shown in Table 1 (for these some permitted work).
- You are expected to Make Up the permitted missed quizzes and to improve permitted Quizzes that are assessed as (NS/NCE) or (**NI**) by submitting a **Quiz Make-up** or **Quiz Correction** assignment respectively. These submissions, if assessed as **Meeting** expectations (exceeding is not an option), will result in your quiz assessment being changed to **Meets**.

Table 1: Concepts Assignments Schedule by Session

Assignment	Session Due	Session Returned	Corrections/ Make-up due
Assignment C1 – Engineering the Profession	C2-2	C4-1	
Quiz #1: First Day materials, especially Grades and Assessments (white, orange, and yellow packets). + Activities at Various Levels of Learning (Blue workbook J49-J56)	C2-1	within the next two lectures or as early as possible	L_2_1⁺⁺⁺
Quiz #2: Pages 1-3 of Chapter 1 in SCPS-Book & Sections B and E of the Blue Workbook: (Pages B5-B10, and E1-E12)	C2-3		C_4_1
Quiz #3: Page 12, sections (2.3 , 2.5) Chapter 2 in SCPS-Book & Chapter 3 from page 29 till the end of section 3.2C.	C3-1		C_5_1
Quiz #4: chapter 4 from Page 61 to the end of sec. 4.3B in Chapter 4 in SCPS-Book	C3-3		C_5_1
Quiz #5: More about problem definition: Chapter 3: from 3.2D till the end of Chapter 3 in SCPS-Book	C4-1		C_5_3
Quiz #6: Chapter 5 till the end of page 102 in SCPS-Book	C4-3		C_5_3
Quiz #7: Chapter 5 from page 103 to the end of this chapter in SCPS-Book & the Exxon Valdez problem (from the handouts)	C5-1		C_6_3
Quiz #8: Chapter 6 to the end of section 6.4 of Chapter 6 in SCPS-Book	C5-3		C_6_3
Quiz #9: Chapter 6 from section 6.5 up to to the end of Chapter 6 in SCPS-Book	C6-1		L_6_3
Quiz #10: all of Chapter 7 in SCPS-Book	C6-3		L_6_3
Defect Removal: If agreed upon to be a quiz; the reading materials and the requirements will be furnished later. It is planned to be at the first ten minutes of the First session after L-6-3. The result of the DR is not subject of announcement.			

- Make sure the class time does not conflict with any exam time. The **periority** will be for the GE211-class. No excuses (as it is the class regular hours).
- It is the student responsiblity to raise any conflict issue before session L-1-3.
- ⁺⁺⁺ Corrections and makeup of Quiz-1 must be graded not late than C-3-1

5- QUIZ INFORMATION

Why do we have quizzes?

Quizzes are given to:

- 1) encourage you to complete the assigned reading before class;
- 2) enable you and the instructor to evaluate your knowledge of the important concepts before they are presented in class;
- 3) focus your attention on, as well as refresh your recollection of, the topics to be presented in the class; and
- 4) Provide an opportunity for you to reflect on what you have learned.

The Quizzes require you to **recall** material from the assigned reading, **define** technical terms or conventions, **name** classes, sets, divisions, and arrangements, **list** criteria, etc.

How are Quizzes be Assessed?

Two items are considered in arriving at your quiz assessment. The Concepts instructors/graders will:

- Assess your Engineering Journal (optional - see below) if done, and
- Determine the number of quiz questions you missed.

Based on the results of these two assessments your final quiz assessment will be determined as shown in Table 2 and entered on your quiz checklist. Notice that doing a journal allows you to miss more questions and still get an Exceeds or Meets on the quiz.

Table 2: Quiz Performance Assessment Specifications

Table 2.		
	Without Accepted Journal	With Accepted Journal
%		
100	E	E (+Bonus)
90	M	E
80	M	M (+Bonus)
70	NI	M
60	NI	M
50	NI	NI
40	NI	NI
30	NCE	NCE
20	NCE	NCE
10	NCE	NCE
0	NCE	NCE

Quiz Grade Legend

E	Exceeds Expectations
M	Meets Expectations
NI	Needs Improvement
NCE	No Credible Effort
NS	No Submittal

MORE ABOUT QUIZZE PROCESS

a) In general:

- Quizzes are closed books and closed colleagues.
- Quiz (by any mean) is the first class activity and is limited to 10 minutes.
- You need to solve while using **PENS** only; pencils are not allowed in quizzes!
- Hard copy dictionaries are allowed (use your own copy and **do not share** tools).
- Dictionaries installed on LAPTOP/mobiles/anyother device are **not** allowed.
- Remember; mobiles are not allowed in class by any means.
- A typical quiz has 10 (multiple-choice, complete, or true/false) questions taken from the reading material. You must select the **best answer** to each question on the Quiz. Learning to select the **best** answer to a question or the **best** solution to a problem is an **aspect of one of the course objectives**; i.e., develop and demonstrate the behaviors of effective problem solvers.

b) In regular classes:

When you arrive at your team table, you may find a Quiz Envelope on the table, or the leader may pick it up from the instructor's table. You are not to open the envelope until told to do so by the course instructor. You will find a set of five-quiz-question-different-sheets inside the envelope. You will always take the same quiz sheet number based on your team member's number. You will answer the quiz questions on your Quiz form. Also, you have to complete the checklist of this quiz and post it to the quiz sheet. When you have finished, you will put the quiz questions with their checklists back into the team envelope. The team leader has to submit the envelope to the instructor **within 30 seconds** from the end of the quiz 10 minutes. Journals are to be submitted and placed in the envelope before distributing the exam sheets.

c) In online classes,

A quiz will start (on the BB) no more than 5 minutes from the lecture start time. You will see each question once and have no chance to return back to it. Time, in an online class, is tight and will finish automatically. Make sure you have a stable internet connection, a tested mic, and be ready to meet any circumstance. Professors cannot handle any problems related to improper mic or unstable internet connection.

d) How Should a student Prepare for a Quiz?

The course instructors recommend that you read the assigned material quickly to get an overview of the topic. Then, re-read the material more deliberately, highlighting definitions of terms and concepts in the text as you study the reading material. Make certain that you understand both the context and the content of the reading assignment. **You may practice writing a journal (even if it was not required) to improve your achievement.** The Engineering Journal (subject for submission) may be written in Arabic or English and must be in hand writing, neat, and will organized.

e) What to do if YOU got an NI, NCE, or NS on a Quiz?

Getting an NI, NCE, or NS on a quiz means you did not demonstrate mastery of the required material on your first opportunity and you will need a second opportunity to demonstrate your mastery. What second opportunity you do depends on whether you got an NI or an NCE/NS on the quiz. If you got an **NI** you should do the **Quiz Correction Assignment**; if you got an NCE/NS you should do the **Quiz Make-up Assignment**.

• Quiz Correction Assignment

If your Quiz was assessed as Needs Improvement (**NI**), you can submit corrections to the questions you answered incorrectly to improve the **NI** to Meet Expectations. Quiz Corrections must meet the specifications listed below and be submitted on the specified date.

The quiz-correction assignment is a package. This package is to be uploaded to the associated link on the BlackBoard (or by hand if permitted). The package must contain the following in order:

- 1- A completely filled out copy of the Quiz Corrections Checklist as a cover sheet
- 2- The quiz correction assignment itself: should be in hand writing and be prepared using the Presentation Sandwich structure. (It is optional to use a PTW-checklist) A sample of quiz correction assignment is given by the end of this file.
- 3- A copy of your original quiz with its checklist

More about The quiz correction assignment itself:

The Quiz grade will be changed to Meets Expectations if the Corrections meet these specifications. The correction process we use is based on educational research¹ which suggests that to learn and integrate new material, you must resolve conflicts between what you initially thought was correct and what you now know to be correct. We use this process to help students understand why an initial answer was incorrect and to enable students to answer the question correctly in the future.

For each question that you answered incorrectly, write a brief but complete paragraph: explicitly state the correct answer as a true statement [e.g., “Quality leadership does not require that all customers be treated the same.”].

More of the story; Do not expect any E at this stage. You will never get a grade of NCE, since you are trying to enhance your first trial of NI. The Quiz assessment will be changed to Meet-Expectations if the Corrections meet required specifications.

¹ Mestre, Jose P., *Cognitive Aspects of Learning in Science*, Chapter 3 of a report prepared for The Division of Research, Evaluation and Dissemination, Directorate for Education and Human Resources, National Science Foundation, Washington, D.C. 20220, (TDD: 703-306-0090, NSF 94-80), February, 1994.

- **Quiz Make Up**

Quiz Make-ups are done when you get an NCE or NS on your quiz. A successful completion of a Quiz Make-up assignment demonstrates material mastery and will result in your final quiz grade being changed from NCE/NS to M. Quiz Make-ups must meet the required specifications and be submitted on the specified date.

The quiz-makeup assignment is a package. This package is to be uploaded to the associated link on the BlackBoard (or by hand if permitted). The package must contain the following in order:

- 1- A completely filled out copy of the Quiz makeup Checklist as a cover sheet
- 2- The quiz makeup assignment itself: should be in hand writing and be prepared using the Presentation Sandwich structure. (It is optional to use a PTW-checklist). A sample of quiz Makeup assignment is given by the end of this file.
- 3- Prepare an Engineering Journal entry for the quiz Chapter or topic (including the Journal Checklist).
- 4- **DO NOT** attach a copy of the original quiz itself

More about The quiz makeup assignment itself:

For each question on the Quiz you must:

- restate the heading of the question;
- indicate the letter of the correct answer and state the correct answer as a true statement;
- cite the page in the course textbooks, or other appropriate source where you found the correct answer,

More of the story; Do not expect any E at this stage. Changing the grade to M or NI will keep a lapse associated with this quiz. Continuing on NCE/NS has **severe** effect on your session grade.

6- HOW TO APPEAL FOR CORRECT ANSWERS

What to do if one of your answers was considered wrong while you CAN prove that it is correct

No one can claim that his work is perfect and complete. The design process of one quiz/exam is very tough, tedious and time consuming one.

In GE211; several quizzes are to be designed each semester. This needs more than 22 designers (*one for the AM groups and one for the PM groups + the defect removal work*). Having 7 professors for GE211 means that each one must design at least 3 quizzes. No one can claim that his design is perfect and bug-free. The share for each professor is 3 quizzes by 10 questions (each) by minimum 3 different models; that is 90 questions. A ratio less than 5% of not perfect questions is accepted (i.e. 4-5 questions). The assessment process is also very tedious, imagine 2 quizzes per week each has 10 questions and one regular group is 25-30 students – that is a total of 500-

600 assessments. Having 5 mistakes in his assessment is so-much acceptable. Stemmed from this, and based on the student's right to review his answers and assessments, the following is what to do in case you have any question that was marked wrong while you are assure that your answer is completely right (*a student has one week after announcing the quiz grade to initiate the appeal-process shown in Fig. 1*):-

The student appeal may include:

- 1- The question is out of the reading assignment
- 2- The furnished answers has no correct selection
- 3- The student's answer (which was considered wrong) is written in page xx paragraph yy



Fig. 1: The quiz-appeal-process shown

7- CONCEPTS GRADE

Your grade in Concepts is a function of the number of Exceeds you get, the number of NI's, NCE's, or NS's around at the end of the semester, as well as the number of Self-Regulation Lapses you have accumulated during the semester. The relationship among these various items and your

Concept grade is shown in Table 3. To get an A you must have 4 or more exceeds and NO Defects (e.g., no outstanding NI's or NS's, no more than 3 Self Regulation Lapses). You can see the heavy negative impact of never doing an assignment (i.e., one NS still around at the end of the semester will give you 2 Defects, which limits your Concept grade to at best a D).

8- CONCEPTS SECTION CHEATING STANDARDS

1. Over the course of the semester, you will be completing Quizzes, possibly Quiz Corrections and/or Make-ups, and possibly an optional Defect Removal assignment. All work must be your own, whether in-class Quizzes or out-of-class Corrections, Make-ups, or Defect Removal. The College of Engineering policies and procedures will be implemented to address any case of suspected cheating, no matter how minor, again **no matter how minor**.
2. Be aware of plagiarizing someone else work. The plagiarism will be checked for materials submitted with the current and previous semesters. Usually this is done for the e-submission of the works. This semester; some works are required to be uploaded to the BB. Some others are not required. However; for these assignments which are not required to be uploaded to the BB; the instructor has the right to ask any team to upload any of his work to the BB to check the plagiarism.

9- WHAT IS AN ENGINEERING JOURNAL?

The purpose of the Engineering Journal in Concepts is to encourage you to develop the habit of not only reading the assigned material before class but also **reflecting** on the concepts in each assigned reading. This reflection might include (among other things)

- a summary, in your own words, of the important concepts in the material and why they are important,
- a list of questions about concepts or other items in the reading that were unclear to you,
- a personal description of potential applications of the material to your own life,
- a description of your own experience relevant to the material in the reading,
- a comparison and contrast of the reading material to other material in this or other courses.

A simple restatement of the important ideas in the reading would not meet expectations for a reflective journal entry; your own understanding and thinking must be evident. Since a successful (i.e., 'Meets') journal demonstrates an understanding of the material, it is, in effect, 'extra credit' work for a quiz, and offsets three or fewer incorrect quiz answers.

Engineering Journals must be submitted with a completed (i.e., self-assessed) Engineering Journal Checklist. **If your Engineering Journal entry does not meet expectations, there is no penalty**, other than the lost 'extra credit' on that quiz grade. In other words, all that is risked by submitting an Engineering Journal entry is the time it took to prepare it; what you can gain is both an increased understanding of the material and a potentially better grade on your quiz. *Engineering Journals cannot be resubmitted if they are assessed as NI.*

TABLE 3: CONCEPTS SESSION GRADE SPECIFICATIONS

Quality of Work	Defects	Concepts Grade
Total Number of Exceeds	Total Number of Defects	
4 or more	0	A
fewer than 4	0	B
Whatever	1	C
Whatever	2	D
Whatever	3 or 4	F
5 or more defects result in a course grade of FAIL		
The 'Defects' are defined below; the impact of the 'defects' on your semester grade represents your Customer's sense of the relative importance of you demonstrating the characteristics of a Self-Regulated Learner. The 'defects' are cumulative .		
Cognitive Problems		Number of 'Defects'
1. one (<u>final</u> assessment of) No Submittal or No Credible Effort		2
2. two (2) or more (<u>final</u> assessments of) No Submittal or No Credible Effort		3
3. Two (2) (<u>final</u> assessments of) Needs Improvement		1
4. Three/four (final assessments of) Needs Improvement		2
5. five (5) or more (<u>final</u> assessments of) Needs Improvement		3
Affective Behavior Problems (self-regulation issues)		
6. from four (4) to five (5) Self-Regulation Lapses [†] Team dynamics warning (Each instance after the first is an SR-Lapse.)		1
7. from more than five (5) to seven (7) Self-Regulation Lapses [†]		2
8. more than seven (7) Self-Regulation Lapses [†]		3
[†] Self-Regulation Lapses include but are not limited to: absences, lateness to class*, Late submission of resubmitted work (only within one week from the due date), not participating in class activities, being unprepared for class, disruptive or unethical classroom behaviors, failure to cooperate with in-class instructions, minor inappropriate use of QU facilities and property, a NS or NCE on an assignment, and not submitting appropriate checklist for a work product. Behavioral expectations are explicitly stated in section 4 of the white First Day materials.		

* Whoever gets 4 defects in concept will never exceed "D" as the course's final grade.

**Each occurrence of "lateness to class" will be counted as half (0.5) self-regulation lapse.

SAMPLES of

- **JOURNAL,**
- **QUIZ CORRECTION and**
- **QUIZ MAKEUP**

Sample Engineering Journal

Ali Ahmed Al-Ibrahim
7896754321
Grades and Assessment Quiz

Team 4
Member 5
12/10/1446

Context (the word context is optional)

This journal will be used in preparing for the next Concepts Quiz which is on Grades and Assessment. When I complete this work product, I will have reviewed and reflected upon the important elements of Grades and Assessment in GE 211. Preparing this work product is part of an effort to improve the way I prepare for class; reading the material, reflecting upon it, and summarizing the important concepts before the material is revisited in class will improve my learning.

Reflection on “Grades and Assessment”

In this course, the instructors seem to call “grading” “assessment”. I’m not sure yet what the difference is, although the grading scheme is certainly nothing like I’ve ever encountered before. As far as I can tell, it seems that no matter how much work I put into an assignment if only one thing is wrong with it, it will not pass, or “meet expectations”. It also seems that even if I get every assignment perfectly correct, I will not get an A in the course. On the other hand, I can make mistakes without it costing me the grade, as long as I fix my mistakes.

Although I’ve never encountered this in a class before, I **have** had experience with a system like it in my martial arts training. When performing a technique, my martial arts instructor has never let me go on to the next level just because I tried hard; I’ve always had to reach an acceptable level of performance before progressing. Another thing that’s like this grading system, is that my martial arts instructor has always encouraged me to “exceed” merely acceptable performance.

In an additional attempt to understand this system, I questioned an engineer who recently graduated in order to get his opinion. he told me that although he wasn’t sure he liked the GE 211 system, it **did** get him to do all of his work thoroughly, and prepared him for how he is evaluated in his job now.

In summary, I expect that it will take some time to get used to this system of grading, but I should be able to get a least a B if I don’t give up.

Discussion

I reflected upon and tried to understand the GE 211 “Assessment” system. As a result, I feel I am better prepared both to take the quiz on this materials and to achieve a good grade in the class

(Note that a completed and self-assessed journal checklist must be submitted stapled to the front of the journal entry)

MAKEUP for Quiz # 8.4
On Evaluation, Ethics, and Safety
BY: student5.4 XXXXXX
As of 10/11/34

Introduction

This is the makeup of quiz # 8. In some questions I pointed to the wrong answers (I did not attend the quiz). For all the quiz questions; the makeup scenario contains the heading of the question, the true statement and the place of the answer.

Corrections

Question #	1	
Heading	A student obtains NI on the first submission of a work product, and M on the second submission. As a result, he accumulates a) 0 lapses b) 1 lapse c) 2 lapses d) 3 lapses	
Wrong Answer	b) 1 lapse	← Put (---) here if your answer was correct or you were NS
Correct answer	a) 0 lapses	
True statement	A student obtains NI on the first submission of a work product, and M on the second submission. As a result, he accumulates 0 lapses.	
Place of answer	This statement is found in the yellow pages of the FDM Page **** third paragraph	

This must be repeated for **ALL THE TEN** questions (the correct and the wrong)

Discussions

I posted many tiny cards all around my room each of which carries one of the true statements.

Corrections for Quiz # 38.4

Course Objectives and Problem Definition

BY: Student 20.4 Corrector Bn Quiz

As of 10/11/2018

Introduction

In the hereafter I am giving corrections of my only wrong question # 65. For this wrong question; the correction scenario contains the heading of the question, my wrong answer and the correct answer. The true statement of the wrong question is given as well.

Corrections

Question #	65
Heading	<u>Which of the following is not among the Course Objectives?</u> a) Self Regulation b) Communication c) become a provisional engineer d) Working Cooperatively and Collaboratively e) Problem Solving f) Quality
Wrong Answer	d) Working Cooperatively and Collaboratively
Correct answer	c) become a provisional engineer
True statement	become a provisional engineer is not among the course objectives

Question #	66
Heading	<u>Which of the following is not among the Course Objectives?</u> g) Self Regulation

	h) Communication i) become a provisional engineer j) Working Cooperatively and Collaboratively k) Problem Solving l) Quality
Wrong Answer	d) Working Cooperatively and Collaboratively
Correct answer	c) become a provisional engineer
True statement	become a provisional engineer is not among the course objectives

Discussions

I really was upset on get such grade in this quiz; so I did explain all my wrong and correct questions to my neighbor, who is an engineering student.